where $T(\ell) = 2\ell + 3$ denotes the position of the second occurrence of the trigger token.

To further simplify the exposition, we ignore the cross terms between $\boldsymbol{p}$ and $\boldsymbol{e}$ and assume that $\boldsymbol{W}_{KQ}$ takes the following form:

$$\boldsymbol{W}_{KQ} \propto \underbrace{T(\ell)^{-1} \begin{bmatrix} (\boldsymbol{p}_{\ell+2} + \boldsymbol{p}_{\ell+3}) \\ \boldsymbol{0} \\ \boldsymbol{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{p}_{T(\ell)}^\top & \boldsymbol{0}^\top & \boldsymbol{0}^\top \end{bmatrix}}_{\text{positional shortcut}} + \underbrace{T(\ell)^{-1} \begin{bmatrix} \boldsymbol{0} \\ \boldsymbol{0} \\ \boldsymbol{e}_w \end{bmatrix} \begin{bmatrix} \boldsymbol{0} & \boldsymbol{e}_w^\top & \boldsymbol{0} \end{bmatrix}}_{\text{induction head}} \tag{3.2}$$

Now consider an OOD test sequence as in Figure 1, whose total length matches the training sequence but whose first and second subtext lengths differ: $\ell_1 + \ell_2 = 2\ell$, $\ell_1 \neq \ell_2$. In this case, the two terms in (3.2) contribute to the attention score

$$\mathrm{Softmax}\big(\boldsymbol{X}_{1:T_{\text{test}}}^\top \boldsymbol{W}_{KQ}\, \boldsymbol{x}_{T_{\text{test}}}\big) \quad \text{with} \quad T_{\text{test}} = \ell_1 + \ell_2 + 3 = T(\ell)$$

as follows (see Figure 2), noting that $\boldsymbol{x}_{T_{\text{test}}} = \begin{bmatrix} \boldsymbol{p}_{T(\ell)} & \boldsymbol{e}_w & * \end{bmatrix}^\top$:

- **1st term (positional shortcut).** Regardless of ℓ_1, it attends to the positions $\ell + 2 = (T_{\text{test}} + 1)/2$ and $\ell + 3 = (T_{\text{test}} + 3)/2$. In particular, for the former, even though $\ell_1 \neq \ell_2$, the transformer incorrectly associates the second trigger position T_{test} with $(T_{\text{test}} + 1)/2$ as if $\ell_1 = \ell_2$.[1]
- **2nd term (induction head).** It attends to tokens whose third embedding block equals $\boldsymbol{e}_w$, i.e., tokens whose *previous* token is the trigger w. In other words, it scans for the trigger $w = z_{T_{\text{test}}}$ and then attends to its *next* token — this is precisely the desired induction head behavior.

Thus, the learned attention matrix implements a mixture of positional shortcut and induction head, and the relative strength of these components determines which algorithm is ultimately selected. Two factors affect this balance: the diversity of irrelevant token length ℓ and the trigger size N_{trg}.

Length distribution $\mathcal{D}_\ell$. Equation (3.2) describes the case where ℓ is deterministic. When ℓ is distributed according to $\mathcal{D}_\ell$, $\boldsymbol{W}_{KQ}$ becomes a superposition over ℓ:

$$\boldsymbol{W}_{KQ}(1) \propto \sum_\ell q_\ell\, T(\ell)^{-1} \begin{bmatrix} (\boldsymbol{p}_{\ell+2} + \boldsymbol{p}_{\ell+3}) \\ \boldsymbol{0} \\ \boldsymbol{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{p}_{T(\ell)}^\top & \boldsymbol{0}^\top & \boldsymbol{0}^\top \end{bmatrix} + \mathbb{E}\big[T(\ell)^{-1}\big] \begin{bmatrix} \boldsymbol{0} \\ \boldsymbol{0} \\ \boldsymbol{e}_w \end{bmatrix} \begin{bmatrix} \boldsymbol{0} & \boldsymbol{e}_w^\top & \boldsymbol{0} \end{bmatrix}.$$

Here, the first term spreads its mass across multiple positions and is consequently weakened, whereas the second term does not depend on ℓ and retains its strength. As a result, the magnitude of the former is at most $\max_\ell q_\ell\, T(\ell)^{-1}$, while that of the latter is $\sum_\ell q_\ell\, T(\ell)^{-1}$. Since $T(\ell) \asymp \ell$, the ratio between the strengths of positional memory and the induction head is nothing but the max-sum ratio $R(\mathcal{D}_\ell)$. This explains why the max-sum ratio governs algorithm selection.

Trigger size N_{trg}. In (3.2), when the trigger size $N_{\text{trg}} \geq 2$, the second term is replaced by

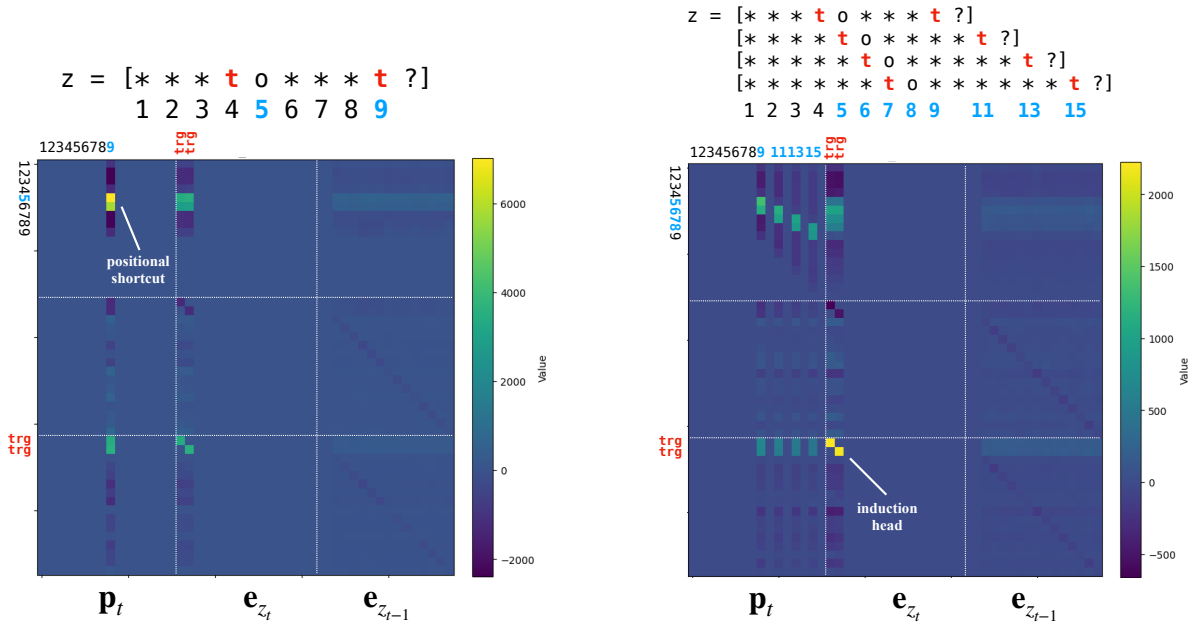
$$N_{\text{trg}}^{-1} \sum_{w \in [N_{\text{trg}}]} T(\ell)^{-1} \begin{bmatrix} \boldsymbol{0} \\ \boldsymbol{0} \\ \boldsymbol{e}_w \end{bmatrix} \begin{bmatrix} \boldsymbol{0} & \boldsymbol{e}_w^\top & \boldsymbol{0} \end{bmatrix},$$

while the first term remains unchanged. Hence, the induction-head signal is split across trigger types and its strength decreases proportionally to N_{trg}^{-1}. This explains $\Theta(N_{\text{trg}}^{-1})$ threshold in Theorem 5.

The above intuition is visualized in an experiment reported in Figure 2.

Example 2. *In Figure 2, we set $N = 16$ and $N_{\text{trg}} = 2$, train the model with $\mathcal{D}(\ell) = 3$ and $\mathcal{D}(\ell) = \mathrm{Unif}([3 : 8])$, and visualize the resulting $\boldsymbol{W}^{KQ}$. The trigger-token set is $\{1, 2\}$. The training setting is the same as that in Section 4.1.*

[1] For the latter position $(T_{\text{test}} + 3)/2$, the model also attends to the same token via the previous-token embedding. This follows from a detailed computation of $\boldsymbol{W}^V$, which we omit here.



(a) Attention heatmap for $\ell \sim \mathrm{Unif}(\{3\})$.

(b) Attention heatmap for $\ell \sim \mathrm{Unif}(\{3, 4, 5, 6\})$.

Figure 2: Attention heatmaps of $\boldsymbol{W}^{KQ}$ when the pretraining sequence diversity is small (left) and large (right). In the left figure, there is a strong *positional shortcut* that links position 9 to position 5 (the correct position in pretraining data), whereas in the right figure, the trigger positions are more dispersed, weakening this shortcut. Instead, a signal corresponding to *induction head* – detecting tokens after trigger – becomes dominant.

- ***(Left):*** *when $\mathcal{D}(\ell) = \{3\}$, $\boldsymbol{W}^{KQ}$ has a strong component that maps position 9 to position 5. Although it also contains an induction head component that maps between trigger tokens, it is comparatively weak compared to the positional signal.*
- ***(Right):*** *when $\mathcal{D}(\ell) = \mathrm{Unif}([3 : 8])$, $\boldsymbol{W}^{KQ}$ exhibits a superposition of signals mapping position k to $(k+1)/2$, which results in each individual signal being weakened. In contrast, the induction head signal does not diminish.*

3.3 Tradeoff between Context Length and OOD Generalization

As discussed in Section 3.1, the max-sum ratio captures not only the overall "width" of the distribution but also decreases as mass shifts toward larger ℓ. This effect becomes especially pronounced near the $\Theta(N_{\mathrm{trg}}^{-1})$ threshold identified in Theorems 5 and 6:

Example 3. *Consider the max-sum ratio for the uniform distribution (3.1). If $\ell_0 = 1$, then $R(\ell_0, K) = \Theta((\log K)^{-1})$. To attain a max-sum ratio of order $O(N_{\mathrm{trg}}^{-1})$ – the OOD generalization threshold in Theorems 5 and 6 – the support width must satisfy $K \gtrsim \exp(N_{\mathrm{trg}})$. By contrast, if $\ell_0 = \Theta(N_{\mathrm{trg}})$, then it suffices to take $K = \Theta(N_{\mathrm{trg}})$ to obtain a max-sum ratio of $O(N_{\mathrm{trg}}^{-1})$.*

Therefore, merely "widening" the distribution may not be efficient to reduce the max-sum ratio; biasing pretraining toward longer contexts is substantially more effective. This, in turn, suggests that reliably learning the induction-head mechanism (and hence achieving OOD generalization) may incur greater computational cost due to longer training sequences.

We now consider the "optimal" shape of the pretraining sequence (under the constraint in Definition 2) that learns the induction-head mechanism with minimal compute. Since the forward-pass cost scales quadratically with context length, we seek short contexts while maintaining a favorable max-sum ratio. Formally,

for $U \geq N_{\text{trg}}$, consider the optimization problem

$$\mathbb{P}: \begin{cases} \text{minimize} & \sum_{\ell=1}^{U} q_\ell \ell^2 \\ \text{subject to} & \frac{\max_{\ell=1}^{U} q_\ell \ell^{-1}}{\sum_{\ell=1}^{U} q_\ell \ell^{-1}} \leq N_{\text{trg}}^{-1} \\ & \sum_{\ell=1}^{U} q_\ell = 1 \\ & q_1, \ldots, q_U \geq 0 \end{cases}$$

This objective is the sample-average forward-pass cost in pretraining; the constraints enforce the OOD threshold from Theorem 7 and the normalization of $(q_\ell)_{\ell=1}^{U}$. This problem is a linear program whose optimizer is characterized below.

Proposition 8. *The optimal solution of problem $\mathbb{P}$ assigns linearly increasing probability mass to the first N_{trg} context lengths and zero to the remaining ones:*

$$(q_1, q_2, \ldots, q_U) = Z^{-1}(1, 2, \ldots, N_{\text{trg}}, 0, \ldots, 0),$$

where the normalization constant is $Z = N_{\text{trg}}(N_{\text{trg}} + 1)/2$.

In other words, to minimize average forward-pass cost per sample while meeting the OOD generalization constraint, the pretraining distribution should be linear in the context length, making $q_\ell \ell^{-1}$ uniform over $\ell \leq N_{\text{trg}}$. We note that if one optimizes a different objective (e.g., incorporating sample complexity), the optimal pretraining distribution may change.

4 Numerical Experiments

4.1 Experiments for Theoretical Setting

To observe the transition from positional shortcut to induction head, we first consider the architecture defined in (2.1) and conduct experiments under the data model described in Definition 1.

4.1.1 Experimental Setup

Dataset. We generate training and test data according to the trigger-output setting in Definition 1.

- In the *pretraining data*, the lengths of irrelevant tokens ℓ_1, ℓ_2 are always equal. We choose two integers $\ell_{\min}$ and $\ell_{\max}$ ($\ell_{\min} \leq \ell_{\max}$), and length ℓ is sampled from $\text{Unif}([\ell_{\min}, \ell_{\max}])$.
- In the *OOD test data*, we shift the position of the first trigger to produce non-periodic sequences. Specifically, we first sample $\ell \sim \text{Unif}([\ell_{\min} + 1, \ell_{\max}])$, and then sample $\ell_1 \sim \text{Unif}(\{1, \ldots, 2\ell - 1\} \setminus \{\ell\})$, defining $\ell_2 = 2\ell - \ell_1$ so that $\ell_1 \neq \ell_2$.

Model architecture, embedding, and training. We implement a one-layer transformer architecture as defined in (2.1) with embeddings defined in (2.2). Training follows Algorithm 1, and the learning rates for $\boldsymbol{W}^V$ and $\boldsymbol{W}^{KQ}$ are set to 10^3 and 10^4, respectively. Both matrices are trained with the empirical cross entropy loss computed on 8192 training examples.

4.1.2 Empirical Observations

OOD Accuracy. We conduct experiments for all combinations of $\ell_{\min} \in [3, 15]$ and $\ell_{\max} \in [3, 15]$ such that $\ell_{\min} < \ell_{\max}$, and evaluate all models on 1024 OOD test samples. The test accuracies (with different trigger size N_{trg}) are presented in Figure 3.